ACS College of Engineering

Department of Cyber Security

Evolving Encryption and AI Driven Cyber Defense for Satellite Communication Networks

Detailed Project Report

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1. Introduction

This project is a cybersecurity framework designed for satellite communication networks. The main idea is to simulate how multiple satellites and a ground station communicate with each other and then secure that communication using modern cyber defense methods. The project does not focus only on encryption. It also checks whether a message is genuine, whether it has been modified, whether it is being replayed by an attacker, and whether the sender is behaving abnormally over time.

In simple terms, the project behaves like a secure mission control system. Satellites send telemetry data, the ground station sends commands, and every packet must pass through a Zero Trust security layer. Zero Trust means that the system does not automatically trust any device, even if that device already belongs to the network. Every packet must prove itself every time.

This project also introduces intelligence into cybersecurity. It uses an AI-based anomaly detector and a trust score system to monitor the behavior of each satellite. If a satellite starts behaving in a suspicious or malicious way, the system automatically reduces its trust score and can eventually isolate it from the network.

2. Problem Being Solved

Satellite communication systems are very important because they support navigation, broadcasting, scientific observation, weather prediction, remote sensing, and defense-related services. These systems are exposed to cyber threats such as replay attacks, spoofing, command injection, packet tampering, impersonation, and unauthorized access.

Many conventional systems still depend on static trust assumptions and fixed cryptographic arrangements. That means if one security component is compromised, attackers may continue to exploit that weakness for a long time. This project tries to solve that problem by introducing evolving encryption keys, continuous verification, AI-based anomaly detection, and automated cyber defense actions.

3. Project Goal

The goal of the project is to create a secure and intelligent satellite communication simulation that can model real packet exchange between satellites and a ground station, protect each packet with strong cryptography, detect abnormal behavior, calculate trust dynamically, isolate suspicious devices automatically, and present all of this clearly on a mission control dashboard.

4. System Overview

The system is made up of five major parts. The first part is the communication simulator, which creates the satellites and the ground station and controls how they exchange commands and telemetry packets. The second part is the cryptographic security layer, which protects every packet before it is accepted. The third part is the AI anomaly detector and the trust engine, which observe the behavior of each node and measure how trustworthy it is. The fourth part is the autonomous defense engine, which decides what action should be taken when risk increases. The fifth part is the dashboard, which gives a live visual view of the system status.

5. Packet Flow and Communication Process

Whenever a satellite sends telemetry or a ground station sends a command, the system creates a packet. That packet contains the sender identity, receiver identity, packet ID, timestamp, nonce, ciphertext, signature, HMAC, satellite position, ephemeral public key, and evolving key epoch. The packet is then passed through the security layer. Only if all checks succeed is the packet accepted. If any check fails, the packet is rejected and the trust score of the sender is reduced.

6. Hybrid Encryption Used in the Project

The project uses hybrid encryption, which means two types of cryptography are combined in one workflow. The actual packet payload is protected using AES-256-GCM. AES is a symmetric encryption method and is efficient for protecting message data. The cryptographic secret used to produce the packet key is established using ECC, specifically X25519 key exchange. This asymmetric step helps two communicating nodes derive a secure shared secret.

Instead of using the same AES key for a long period, the project derives a new packet-level key using the ECC shared secret and packet-specific context such as packet ID, timestamp, satellite position, and key epoch. This makes the encryption much stronger because each packet effectively gets its own derived key context.

7. Evolving Cryptographic Keys

One of the most important ideas in this project is evolving encryption keys. The key is not static. It changes as packets change. The project uses HKDF, which is a standard key derivation method, to derive fresh AES and HMAC keys for packets. The derivation includes the shared ECC secret and contextual values such as packet ID, timestamp, orbital position, and evolving epoch.

This improves security because even if one key is exposed, it does not automatically reveal future packets. It also makes replay attacks and long-term key compromise much harder. This idea suits

satellite systems well because orbital state and timing are always changing.

8. Authentication and Integrity Protection

Encryption alone is not enough. The project also proves that the sender is genuine and that the message was not changed in transit. For authentication, the project uses Ed25519 digital signatures. These signatures prove that a valid sender created the packet. For message integrity, the project uses HMAC-SHA256. If either the signature or the HMAC check fails, the packet is rejected immediately.

9. Replay Protection

Replay attacks occur when an attacker captures a valid message and sends it again later. To stop that, the project uses timestamps, nonces, packet IDs, and a nonce cache with expiration. If a packet is too old or if the nonce has already been used, the system detects that and blocks the message.

10. Orbit-Aware Security

This project is not a generic computer network simulator. It has a satellite-specific design. Every satellite has a simulated orbital phase and a calculated position. That position changes over time. The position is used both for communication realism and for encryption key derivation. This means the security layer is aware of mission context rather than treating the system like an ordinary terrestrial network.

11. Network Realism Features

To make the simulation more realistic, the system includes latency, jitter, packet loss, bit errors, clock drift, visibility windows, and ground station handovers. These features reflect the fact that real satellite communication is affected by delay, link interruptions, and varying quality conditions. By including these factors, the project becomes more meaningful and less artificial.

12. AI-Based Anomaly Detection

The project includes an unsupervised AI anomaly detector. This means the AI does not need a fully labeled training dataset before it starts. Instead, it gradually learns what normal communication looks like and then measures how different new behavior is from that baseline.

The AI considers features such as sender role, payload type, receiver type, battery level, temperature, and timing patterns between packets. It calculates an anomaly score and a confidence level. If the confidence is high, the trust engine treats that behavior as suspicious.

13. Dynamic Trust Score System

Every node in the system has a trust score. Good behavior slightly increases trust, and suspicious behavior decreases it. The system also considers the role of the node. For example, a ground station is more critical than a regular satellite node, so its trust impact is handled more carefully.

The trust system moves nodes through different operational stages: healthy, monitor, throttle, quarantine, and isolated. This gives more realistic control than a simple allowed or blocked model.

14. Autonomous Cyber Defense

The project contains an autonomous defense engine that does not just detect problems but also responds automatically. Depending on the threat level, it can raise observation level, rate-limit communication, quarantine a device, or fully isolate it and revoke its identity. This is especially important for high-value systems such as satellite communication where fast response matters.

The system also measures defense KPIs such as mean time to detection, mean time to containment, mean time to recovery, containment rate, and recent defense actions. These metrics help show whether the defense engine is effective.

15. Attack Simulation Features

The project can simulate multiple cyber threats. It includes command injection, replay attacks, satellite impersonation, and corrupted packet conditions. These scenarios help prove that the security layer, anomaly detector, trust engine, and defense system are all working together.

16. Mission Control Dashboard

The dashboard is a live visual interface that shows the status of the network. It displays satellites, trust score, AI risk, operational stage, blocked nodes, alerts, defense actions, and performance KPIs. This is important because it helps both technical and non-technical users understand what is happening inside the system during a demonstration.

17. Deployment and Demonstration Support

The project supports local execution, external data ingest, a backend API, a worker service for telemetry injection, GitHub Pages for a free dashboard deployment, and tunnel-based backend exposure for public demos. This makes the project easy to present even without paid cloud services.

18. Work Completed So Far

The project already includes the simulator, hybrid encryption, authentication, replay protection, AI anomaly detection, trust scoring, autonomous defense logic, dashboard visualization, and a demo-friendly deployment path. At the same time, full academic evaluation, benchmarking, and broader real-world integration are still future work. This is why the project is correctly positioned as a progressing prototype rather than a fully finished product.

19. Strengths of the Project

A major strength of this project is that it combines cryptography, Zero Trust verification, orbital context, AI anomaly detection, trust-aware control, autonomous cyber defense, and live visualization in one integrated framework. Many projects focus on only one of these areas, but this project connects them into a unified system for satellite cybersecurity.

20. Limitations

The project is still a simulation-based prototype and is not flight-certified mission software. The AI model is lightweight and designed for academic demonstration rather than industrial deployment. The orbital model is simplified and used mainly for secure context simulation instead of precise astrodynamics.

21. Future Scope

The project can be extended by integrating real telemetry feeds, using more realistic orbital mechanics, adding larger-scale constellation testing, benchmarking multiple AI models, improving packet-level crypto inspection, and preparing the work in a research-paper format. These extensions can make the project stronger academically and practically.

22. Conclusion

This project presents a strong and modern approach to satellite communication security. It does not treat cybersecurity as only encryption or only anomaly detection. Instead, it builds a complete

pipeline where every packet is verified, every node is monitored, and suspicious behavior is handled automatically.

In simple terms, this project creates a smart secure communication environment for satellites. It protects data, checks identities, watches behavior, reacts to attacks, and presents everything clearly on a dashboard. That makes it a meaningful and technically strong final-year project in the field of Cyber Security.